

Mode C RAM Read Timing Issue

Test conditions and waveform

MODE_C; length=4; scale_shift=0; round_en=0; signed_out=1. Source 0x80000000; destination 0. RAM contains 1, 2, 3, 4 at source offsets 0, 4, 8, 12.

Expected: four reads, one write of 0x04030201 at address 0.

Result: The basic test writes the expected 0x04030201 to the scratchpad output, but the RAM request sequence repeats the first address and accesses an additional address. The simulation stops with RAM_UNINITIALIZED.

The responder uses zero wait states with no CPU competition. It samples requests with input #1step and drives HRDATA after the edge with output #0, for sampling at the next rising edge.

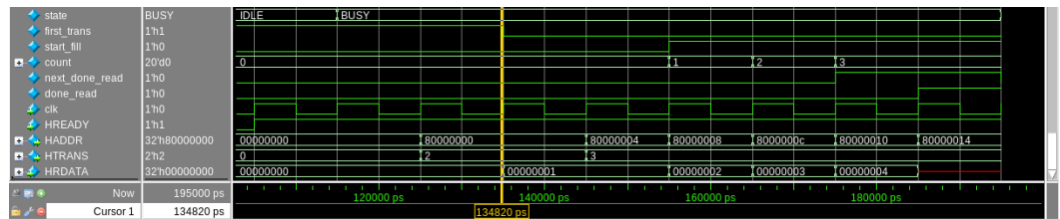


Table 1 Requests and returned data

All signal values below are before the rising edge. Grant and effective HREADY are high at these edges.

Edge ns	HADDR	HTRANS	Responder action at edge	HRDATA before edge	DUT data handling at edge
125	00000000	IDLE	No valid request	0	No capture
135	80000000	NONSEQ	Accept first request; drive 1 after edge	0	No capture
145	80000000	NONSEQ	Accept duplicate; drive 1 after edge	1 from 135 ns	First response skipped
155	80000004	SEQ	Accept request; drive 2 after edge	1 from 145 ns	Capture byte 01
165	80000008	SEQ	Accept request; drive 3 after edge	2	Capture byte 02
175	8000000C	SEQ	Accept request; drive 4 after edge	3	Capture byte 03
185	80000010	SEQ	Uninitialized address causes fatal	4	Scratchpad monitor records 04030201

Table 2 Register transitions and address updates

Each arrow means before edge → after edge.

The count register saturates at length−1, so its final value of 3 is not a total-read count.

Edge ns	first_trans	start_fill	count	done_read	After edge and significance
125	1 → 1	0 → 0	0 → 0	0 → 0	HADDR=80000000; HTRANS=NONSEQ.
135	1 → 0	0 → 0	0 → 0	0 → 0	First address and NONSEQ are loaded again.
145	0 → 0	0 → 0	0 → 0	0 → 0	HADDR=80000004; HTRANS=SEQ. First returned 1 was skipped.
155	0 → 0	0 → 1	0 → 1	0 → 0	HADDR=80000008. Capture starts with the duplicate response.
165	0 → 0	1 → 1	1 → 2	0 → 0	HADDR=8000000C.
175	0 → 0	1 → 1	2 → 3	0 → 0	HADDR=80000010; next_done_read rises to 1.
185	0 → 0	1 → 1	3 → 3	0 → 1	Extra request sampled; fatal occurs. HADDR advances to 80000014 in the waveform.

Code A First address is loaded twice

```

146 if (state == BUSY) begin
147     if (HREADY) begin
148         if (first_trans) begin
149             next_HADDR = src_addr;
150             next_HTRANS = AHB_NONSEQ;
            // ... remaining branches omitted

            // Clocked register assignments, noncontiguous excerpts
222 HADDR  <= next_HADDR;
225 HTRANS <= next_HTRANS;
250 first_trans <= (HTRANS == AHB_NONSEQ) ? 0 : first_trans;

```

At 125 ns, HTRANS loads NONSEQ, but first_trans is evaluated using the old IDLE value and stays high. At 135 ns, first_trans clears; HADDR still loads the first address computed before that edge. The next address is not registered until 145 ns, so both 135 ns and 145 ns accept the first address.

Code B The first returned value is skipped

```
251 start_fill <= (HTRANS == AHB_SEQ) ? 1 : start_fill;
    // Inside the enabled BUSY data-processing branch
256 if (start_fill || (HTRANS == AHB_SEQ)) begin
257     if (start_fill || (HTRANS == AHB_SEQ)) begin
258         count <= (count == length-1) ? count : count + 1;
        // ... other updates omitted
        // Mode C capture inside the outer condition, line 272
        quant[count_bytes*QUANT_DIM +: QUANT_DIM]
            <= quant_fn(HRDATA, scale_shift, signed_out, round_en);
```

At 145 ns, HRDATA=1, but pre-edge HTRANS=NONSEQ and start_fill=0, so capture is disabled. At 155 ns, pre-edge HTRANS=SEQ and capture is enabled. The captured 1 belongs to the repeated request accepted at 145 ns, not the original request at 135 ns.

Code C New addresses continue before completion is registered

```
    // Line 109, wrapped for readability
    assign next_done_read = (count == length-1)
        && HREADY && (state == BUSY);
226 done_read <= next_done_read;

141 next_HADDR = HADDR;
144 next_HTRANS = HTRANS;
    // After the first_trans branch
151 end else if (!done_read) begin
152     next_HADDR = src_addr + (burst << (COLS_WIDTH+2))
        + (beat << 2) + 4;
    // Line 153, wrapped for readability
    next_HTRANS = (beat == NUM_BEATS-1)
        ? ((count == length-1) ? AHB_IDLE : AHB_NONSEQ)
        : AHB_SEQ;
154 end
```

At 175 ns, count reaches 3 and next_done_read rises, but done_read remains 0 until 185 ns. Address 80000010 is already active and is sampled at 185 ns. Also, when done_read is high in BUSY, this branch does not explicitly replace the default retained HTRANS with IDLE.